Virtual Worlds

Programming Assignment #2

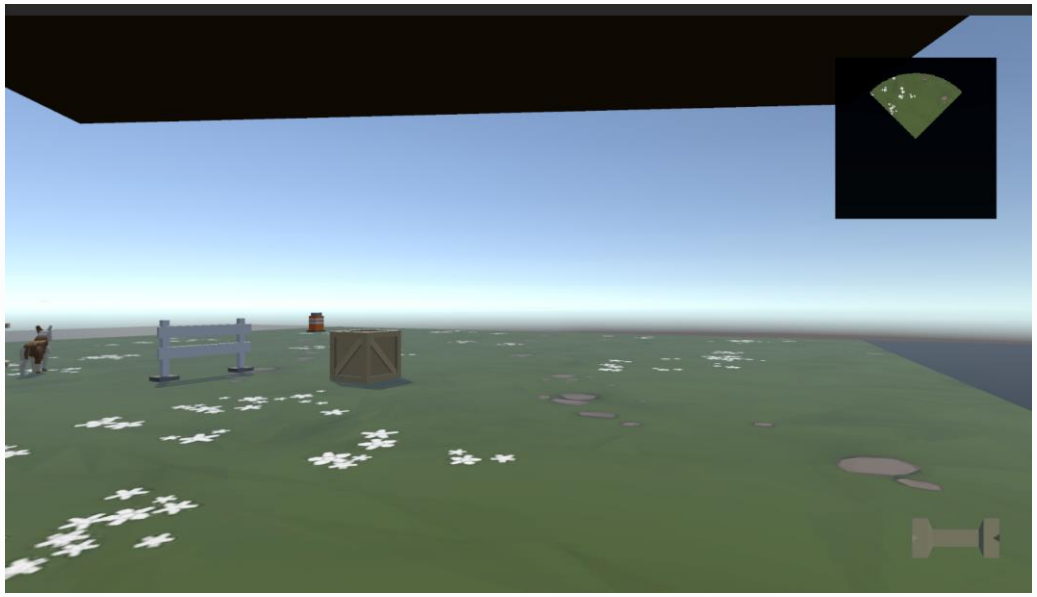
Kang Hoon Lee

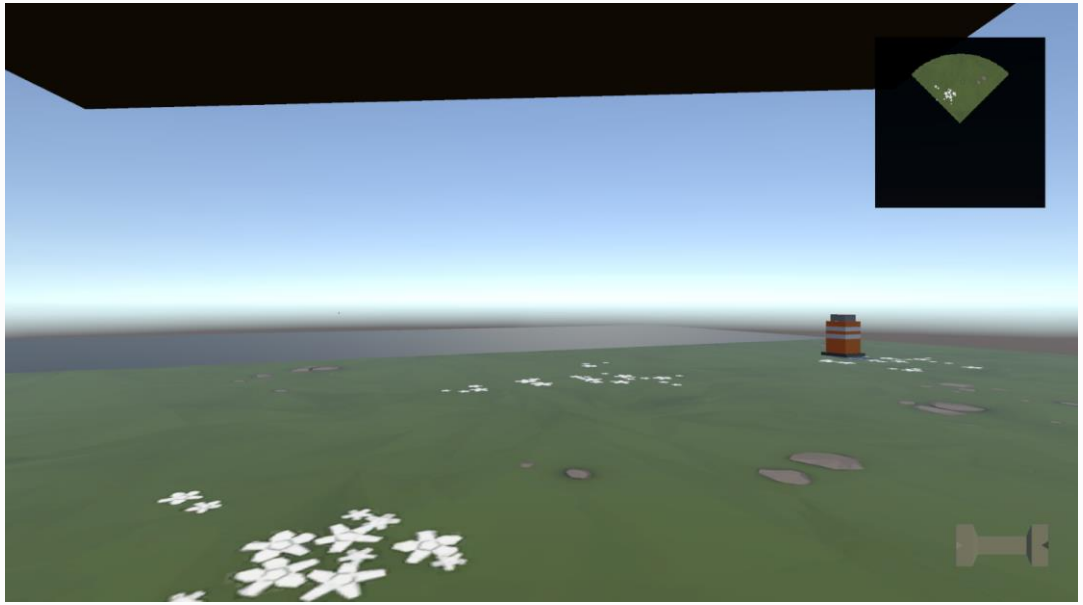
Department of Software
Kwangwoon University

Easy 1: First Person View Movement

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- Create a first-person view that allows moving around. It should enable movement in all directions (up, down, left, right) and allow looking around the surroundings using the mouse within a certain range.





First-person view and movement

Easy 2: Setting up Obstacles

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- Place obstacles at random positions. There should be at least 3 different types of obstacles, and characters and animals should not be able to pass through these obstacles.

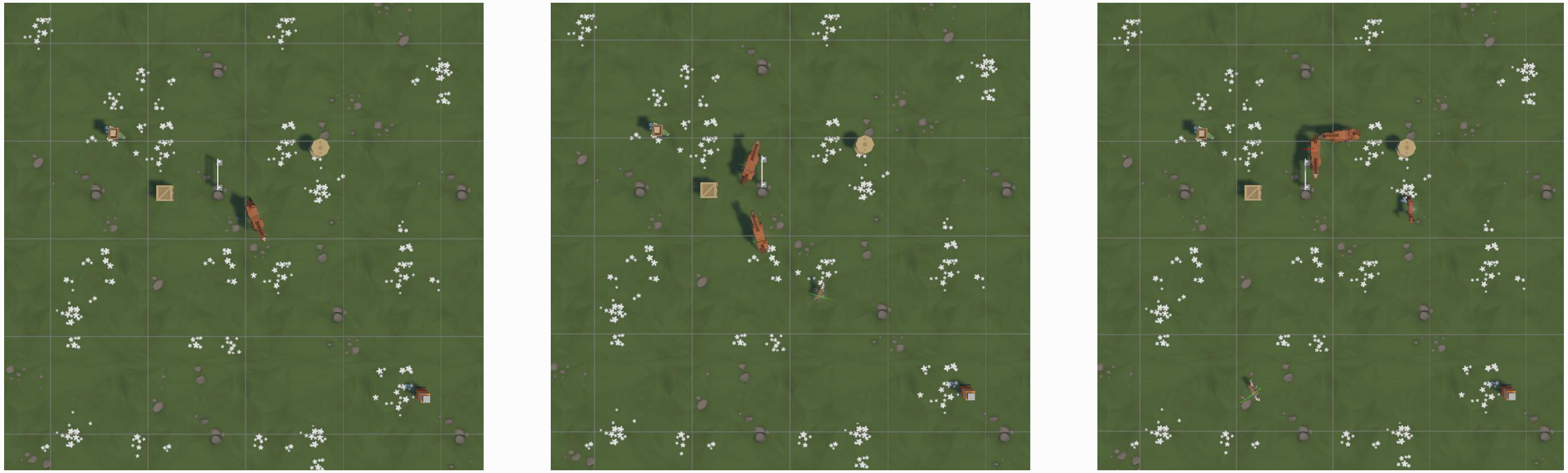


Example of obstacles

Easy 3: Animal Spawn

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- Every 5 seconds, 2 random animals spawn at random locations. The animal types must be fox, horse, and dog.

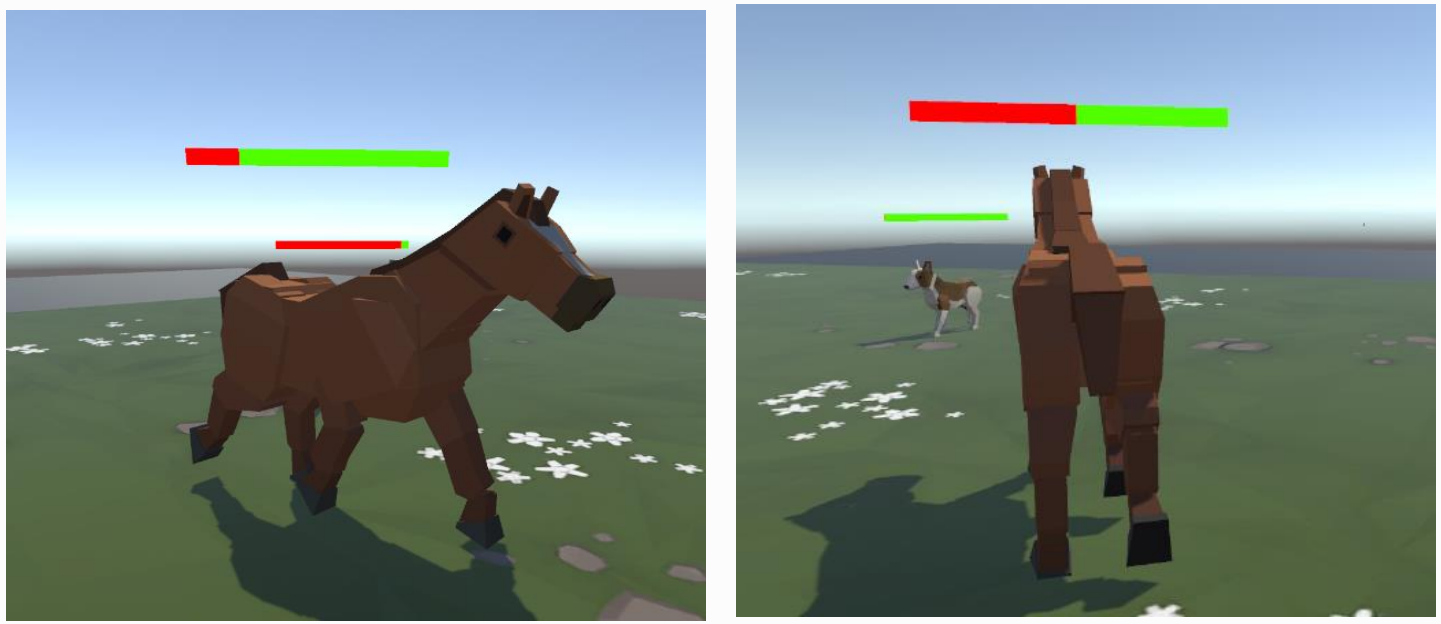


Animal spawn example

Medium 1 : Animal Hunger Bar

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- Display a “hunger bar” on top of each of the animals. When animals are born, their hunger bar start out full. Each time animal eats food, the hunger bar fills up a little. They disappear after their hunger bars decrease all. Hungerbar must always look at the camera.



Hunger Bar decreases over time and always look at the camera

Expert 1: Move freely

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- Animals move around freely. The animal rotates and moves smoothly. But it must not pass through obstacles or the player.





Animal Move Example

Hard : Feeding Animals by Hand

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- Feed the animal that the character is facing (the one at the center of the screen). You must provide the correct type of food depending on the animal. You can change the selected food type by right-clicking, and the currently selected food is displayed in the bottom right corner. Dogs should be given bones, foxes should be given apples, and horses should be given carrots to satisfy their hunger. When eating, the animal should play an eating animation. During this time, the animal must not move. Also, when food is given, it should fly toward the animal.



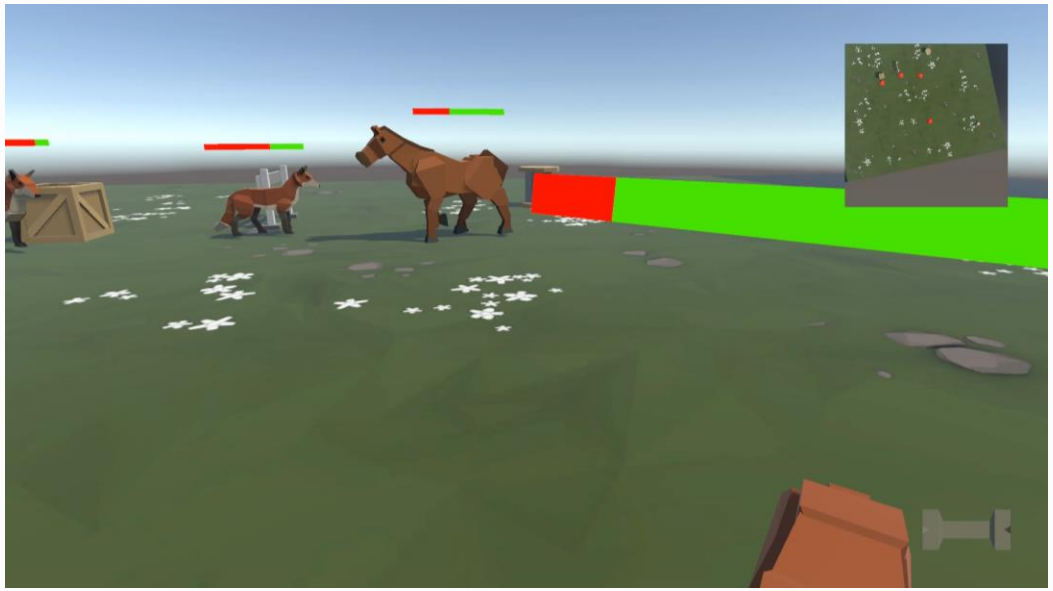


This is a scene showing an animal that has been properly fed eating food. The current food content is displayed in the lower right corner.

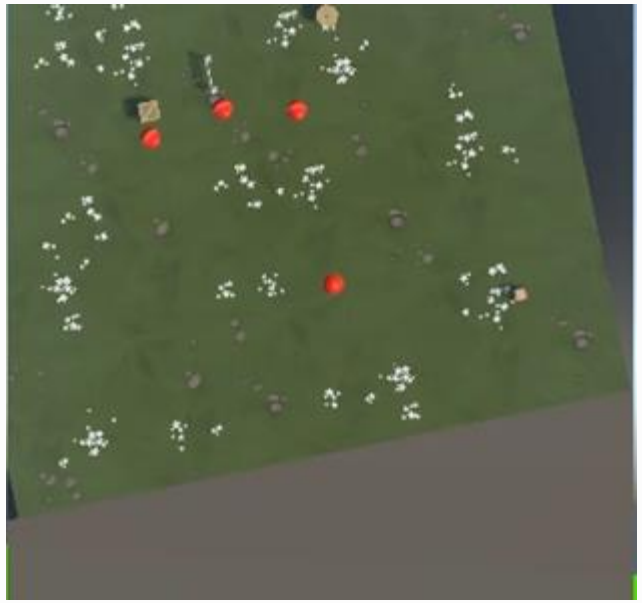
Easy 4: Mini-map 1

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- Implement a mini-map that is displayed in the upper right corner. The player is positioned at the center of the mini-map, which rotates based on the character's movement. On the mini-map, the character is always facing upward. The player is not visible on the mini-map, and animals are shown as red circles.



Explain of mini-map



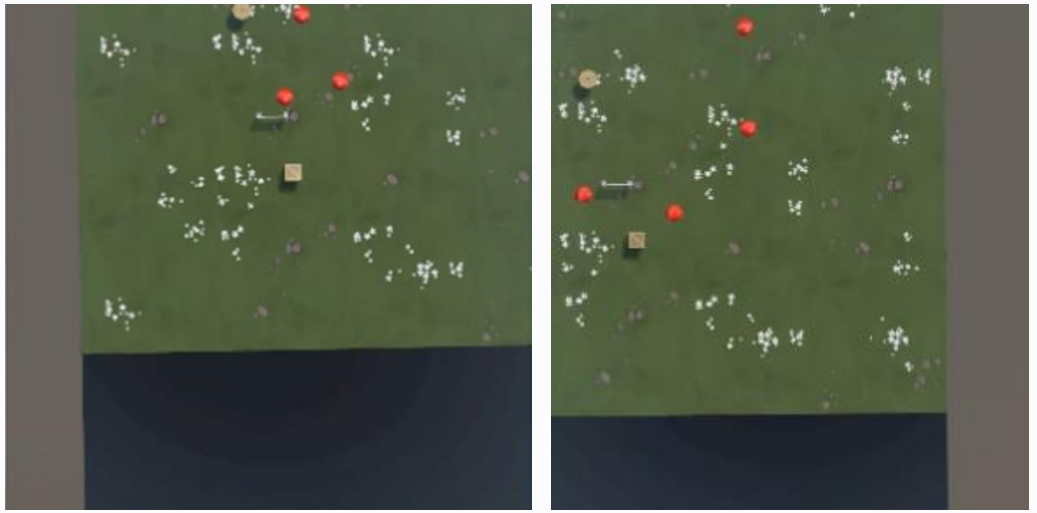
Medium 2: Mini-map 2

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- If the previously implemented mini-map rotated along with the character, now create a mini-map that follows the character's movement while maintaining a fixed orientation. The character should remain fixed at the center of the map and can face different directions on the mini-map. Pressing the R key should toggle between the two mini-map modes.



Rotating Mini-map (Previous Version)

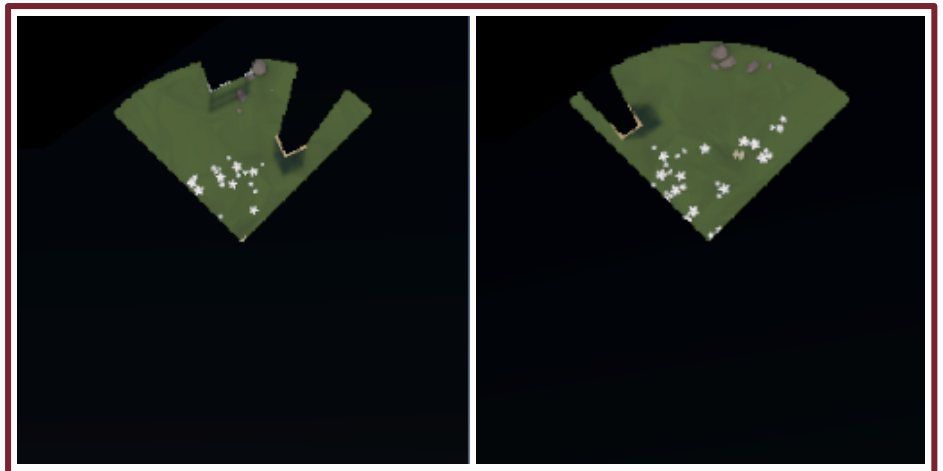


Fixed Mini-map (Current Version)

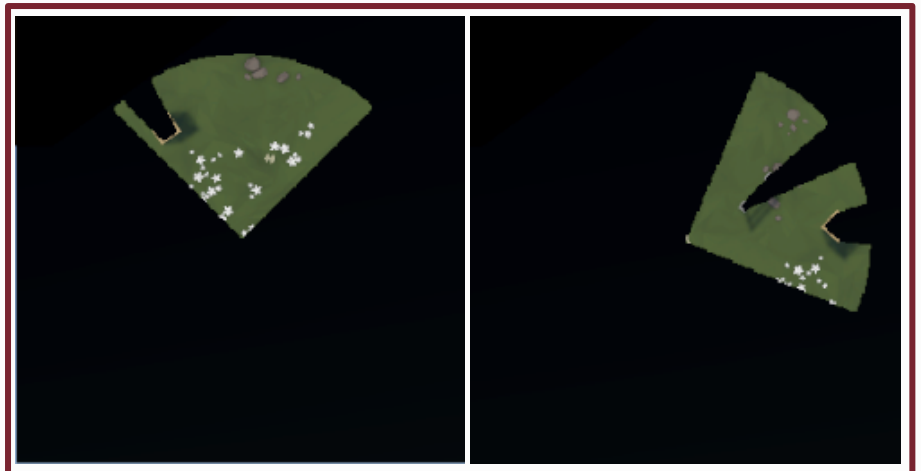
Expert 2: Mini-map 3

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- Create a mini-map that displays only a 90-degree field of view (FOV) in the direction the player is facing. For this task, use the mesh creation, which was covered in the first assignment, along with the masking functionality. If there are obstacles in the player's line of sight, make the area behind those obstacles invisible.



FOV (Field of View) - Rotated Version



FOV (Field of View) - Fixed Version



Mesh example

Submission

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- **Report**
  - Title page
    - Course title, submission date, affiliation, student ID, full name
  - Fulfill table
    - A table to write down what you are satisfied with
    - An example is attached on the next page.
  - For each requirement, explain how you fulfilled it
    - Do not just dump the entire code
    - It's okay to copy snippets of your code to complement written description
    - ❖ Capture and attach a screen shot for each result
  - Conclude with some comments on your work
    - Key challenges you have successfully tackled
    - Limitations you hope to address in the future
  - Add a link to a video that runs the program.
    - Please take a video so we can clearly see what you achieved.
    - When you click the link, the video should run without problems.

Submission

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☐ **Example of Fulfill Table**

Easy 1	First Person View Movement	O
Easy 2	Setting up Obstacles	O
Easy 3	Animal Spawn	O
Easy 4	Mini-map 1	O
Medium 1	Animal Hunger Bar	O
Medium 2	Mini-map 2	X
Hard	Feeding Animals by Hand	O
Expert 1	Move freely	X
Expert 2	Mini-map 3	X

Submission

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- **Compress your export file and report into a single *.zip file**
  - Export File
    - Export > Export Files
      - ❖ Once you have imported the provided unitypackage file, it will be in the top left menu.
    - The format of the export file is " packageFile.unitypackage".
  - Report
    - A single “report.pdf” file
    - ❖ You should convert your word format (*.hwp, *.doc, *.docx) to PDF format (*.pdf) before zipping
  - Name your zip file as your student ID
    - ❖ ex) 20xxxxxxxx.zip

- **Before submitting, import the exported file into an empty project and check that it works well before submitting.**

- **Upload to homework assignment in KLAS**

- **Due at 5/21 (Wed), 11:59 PM**